

SEPTEMBER 2026 - MARCH 2027

Four Golden Portfolio Projects

A progressive evidence plan that demonstrates junior automation and controls competence without overstating field experience

DESIGN PROMISE

Each project is broad enough to prove competence, yet scoped tightly enough to finish and explain in an interview.

4 recruiter-readable projects

From discrete control to integrated SCADA

Every claim backed by tests and documents

Simulation / Training Project labels required

Prepared for Emmanuel Januarie

Study period: 7 September 2026 to 31 March 2027

Workload: 17 hours per full week

Why these four projects

Together, the projects cover the technical arc found in junior automation and control roles: electrical signal paths, PLC logic, modes and sequences, motors and drives, analog instrumentation, process control, HMI/SCADA, industrial networks, troubleshooting, testing and documentation.

GOLDEN RULE

A smaller finished project with traceable tests is stronger than a large unfinished build. Every screenshot must be explainable, every simulation labelled, and every unperformed test marked NOT TESTED.

Portfolio coverage matrix

Competence	P1	P2	P3	P4
Discrete I/O and Ladder	High	Med	Low	High
Modes, sequences and faults	High	High	Med	High
Motors and VFDs	Med	High	Med	High
Instrumentation and 4-20 mA	Low	High	Med	High
HMI/SCADA and alarms	Med	High	High	High
Industrial networks	Low	Low	High	High
Troubleshooting and testing	High	High	High	High
Engineering documentation	High	High	High	High

Universal project release gate

- ☐ Scope, assumptions, exclusions and safety boundaries are written
- ☐ Architecture, I/O/tags and acceptance criteria agree
- ☐ Normal, stop, fault and recovery behavior is tested
- ☐ Expected and actual results are recorded
- ☐ At least three evidence-based fault reports exist
- ☐ Backups, versions and change log are present
- ☐ README can be understood in under two minutes
- ☐ Demo can be delivered without hidden setup or unsupported claims

Project 01

Smart Conveyor Sorting Cell

RECRUITER PITCH

A two-zone conveyor cell that detects, counts and transfers products with clear operating modes, permissives, feedback and fault recovery.

BUILD WINDOW	LABEL
21 September - 1 November 2026	Simulation / Training Project

COMPETENCIES ☐ 24 VDC control concepts ☐ PLC I/O and scan cycle ☐ Ladder Logic ☐ Modes and command ownership ☐ Timers, counters and one-shots ☐ HMI diagnostics ☐ FAT and troubleshooting	SYSTEM SCOPE ☐ Two conveyors, two photoeyes, two motor commands and running feedback ☐ Auto, Manual and Local/Remote simulation ☐ Start permissives, overload trips, jam timing and downstream-ready logic ☐ Count target, batch-complete state and controlled reset ☐ HMI overview, faceplates, alarms and sequence-state display
---	--

Acceptance criteria

- ☐ The system starts only when written permissives are true.
- ☐ Mode and command ownership are visible and deterministic.
- ☐ Normal operation, stop, fault and recovery match the control narrative.
- ☐ Bad signal or communications quality is visible and cannot look normal.
- ☐ Each critical fault produces a defined safe response and useful diagnostic.
- ☐ Documents, PLC/HMI tags and test records remain traceable by revision.

Project 01 build plan

Window	Stage	Required output
21 Sep-4 Oct	Design	Process sketch, 12-point I/O list, tag standard and signal trace
5-11 Oct	Logic library	Start/stop, timer, counter, one-shot and comparison routines
12-25 Oct	Integrate	Modes, sequence, HMI, faults and cause-and-effect
26 Oct-1 Nov	Prove	FAT, defect correction, recovery tests and recorded demo

REQUIRED FAULT INJECTIONS	EVIDENCE PACKAGE
☐ Sensor LED on but PLC input off	☐ README with architecture and assumptions
☐ Repeated counts from a maintained sensor	☐ I/O list and tag standard
☐ Downstream Ready false	☐ Control narrative and sequence chart
☐ Motor command true but feedback missing	☐ Commented PLC logic
☐ Overload trip and reset-before-correction attempt	☐ HMI screenshots
☐ Communication loss and stale HMI indication	☐ Cause-and-effect matrix
	☐ FAT with expected/actual results
	☐ Three fault reports
	☐ Two-minute demo

Project 01 demonstration and scoring

Demonstration script

1. Explain the scan and one signal path
2. Run Auto cycle
3. Switch mode and show ownership
4. Inject a jam fault
5. Recover only after correction
6. Show evidence and limitations

Golden differentiators

- ☐ First-out fault banner
- ☐ State/transition diagram beside live sequence state
- ☐ A before/after diagnosis screenshot pair
- ☐ A recruiter-readable one-page architecture

Category	Weight	Release question
Correct function	25%	Does normal operation match the written narrative?
Fault behavior	25%	Are faults isolated, protected and recoverable?
Traceability	20%	Do drawings, tags, logic and tests agree?
Operator clarity	15%	Are mode, quality, alarms and actions obvious?
Explanation	15%	Can the work be explained honestly without notes?

RELEASE GATE

Target at least 80/100, with no category below 60 percent. A safety, ownership or data-quality defect blocks release even if the total score passes.

Project 02

Automated Batch Tank and Dosing Station

RECRUITER PITCH

A simulated fill, mix, dose and drain process combining motor/VFD control, instruments, 4-20 mA scaling, valves, alarms and basic feedback control.

BUILD WINDOW	LABEL
2 November 2026 - 3 January 2027	Simulation / Training Project

COMPETENCIES	SYSTEM SCOPE
☐ Motors and VFD interface ☐ Sensor selection ☐ 4-20 mA scaling ☐ Analog validation ☐ Process sequences ☐ Valves and fail state ☐ Alarms, trends and PID awareness	☐ Inlet valve, outlet valve, mixer motor and dosing pump ☐ Level and flow transmitters with simulated 4-20 mA signals ☐ Auto batch sequence with hold, abort and controlled restart ☐ Manual/Automatic level behavior; PID awareness without claiming tuning expertise ☐ Alarm priorities, High-High trip, discrepancy alarms and trend diagnostics

Acceptance criteria

- ☐ The system starts only when written permissives are true.
- ☐ Mode and command ownership are visible and deterministic.
- ☐ Normal operation, stop, fault and recovery match the control narrative.
- ☐ Bad signal or communications quality is visible and cannot look normal.
- ☐ Each critical fault produces a defined safe response and useful diagnostic.
- ☐ Documents, PLC/HMI tags and test records remain traceable by revision.

Project 02 build plan

Window	Stage	Required output
2-15 Nov	Actuation	Motor/VFD map, faceplate, command/feedback tests
16-29 Nov	Measurement	Instrument selection, list, ranges and failure states
30 Nov-13 Dec	Analog	Loop diagrams, scaling, quality and alarm tests
14 Dec-3 Jan	Process	Sequence, valves, alarms, trends, FAT and demo

REQUIRED FAULT INJECTIONS	EVIDENCE PACKAGE
☐ Drive left in LOCAL	☐ P&ID and process description
☐ Frozen level value	☐ I/O and instrument lists
☐ Noisy analog signal	☐ Loop diagrams and scaling worksheet
☐ Wrong PLC scaling range	☐ PLC logic and HMI screens
☐ Outlet valve command/position discrepancy	☐ Alarm rationalization table
☐ High-High trip during filling	☐ Trend screenshots with event explanation
	☐ FAT and fault reports
	☐ Three-minute demo

Project 02 demonstration and scoring

Demonstration script

1. Trace level from process to HMI
2. Run one complete batch
3. Show scaling at 0/50/100 percent
4. Inject valve discrepancy
5. Explain trend and protective action
6. Show controlled restart

Golden differentiators

- ☐ Animated but restrained process overview
- ☐ Three-point scaling proof
- ☐ Alarm cause-response-consequence table
- ☐ Trend-based root-cause explanation

Category	Weight	Release question
Correct function	25%	Does normal operation match the written narrative?
Fault behavior	25%	Are faults isolated, protected and recoverable?
Traceability	20%	Do drawings, tags, logic and tests agree?
Operator clarity	15%	Are mode, quality, alarms and actions obvious?
Explanation	15%	Can the work be explained honestly without notes?

RELEASE GATE

Target at least 80/100, with no category below 60 percent. A safety, ownership or data-quality defect blocks release even if the total score passes.

Project 03

Plantwide SCADA and Industrial Communications Lab

RECRUITER PITCH

A small multi-device control network that proves tag mapping, data quality, alarms, trends, commands, timeouts, access control and communications recovery.

BUILD WINDOW	LABEL
4 January - 7 February 2027	Simulation / Training Project

COMPETENCIES	SYSTEM SCOPE
☐ HMI/SCADA usability ☐ IPv4 and subnet basics ☐ Modbus TCP mapping ☐ OPC UA awareness ☐ Read/write permissions ☐ Watchdogs and stale data ☐ Backup, restore and change control	☐ PLC, HMI, SCADA/client, switch and simulated VFD/instrument nodes ☐ IP schedule and protocol map ☐ At least 20 integrated tags with source, data type, scale, access and quality ☐ Alarm/event and diagnostic trend views ☐ Loss-of-comms behavior, local control philosophy and recovery timing

Acceptance criteria

- ☐ The system starts only when written permissives are true.
- ☐ Mode and command ownership are visible and deterministic.
- ☐ Normal operation, stop, fault and recovery match the control narrative.
- ☐ Bad signal or communications quality is visible and cannot look normal.
- ☐ Each critical fault produces a defined safe response and useful diagnostic.
- ☐ Documents, PLC/HMI tags and test records remain traceable by revision.

Project 03 build plan

Window	Stage	Required output
4-10 Jan	HMI design	Hierarchy, overview, faceplates, alarm and trend views
11-24 Jan	Network	Architecture, IP plan, Modbus map and layered tests
25-31 Jan	SCADA	Tags, alarms, trends, access and recovery
1-7 Feb	Fault pack	Ten network/system faults, change log and retest

REQUIRED FAULT INJECTIONS	EVIDENCE PACKAGE
☐ Wrong subnet	☐ Network architecture and IP schedule
☐ Wrong Modbus register	☐ Protocol/tag map
☐ Wrong data type or byte order	☐ SCADA screen pack
☐ Read-only command tag	☐ Alarm and trend tests
☐ PLC offline	☐ Command-access test
☐ Stale value without quality indication	☐ Loss/recovery timeline
☐ Timeout too long	☐ Backup/restore checklist
☐ Incorrect scale	☐ Ten fault reports
☐ Duplicate tag name	☐ Three-minute demo
☐ Restore from wrong version	

Project 03 demonstration and scoring

Demonstration script

1. Show architecture and tag lifecycle
2. Prove a normal data exchange
3. Break one mapping
4. Diagnose by layer
5. Break communications
6. Show stale indication and safe local behavior
7. Restore and time recovery

Golden differentiators

- ☐ Layer-status diagnostic page
- ☐ Single tag traced across all layers
- ☐ Comms-loss event timeline
- ☐ Access-control test showing allowed and denied actions

Category	Weight	Release question
Correct function	25%	Does normal operation match the written narrative?
Fault behavior	25%	Are faults isolated, protected and recoverable?
Traceability	20%	Do drawings, tags, logic and tests agree?
Operator clarity	15%	Are mode, quality, alarms and actions obvious?
Explanation	15%	Can the work be explained honestly without notes?

RELEASE GATE

Target at least 80/100, with no category below 60 percent. A safety, ownership or data-quality defect blocks release even if the total score passes.

Project 04

Integrated Water Treatment Automation System

RECRUITER PITCH

The flagship: a simulated raw-water, treatment, filtration and storage process that unifies PLC, instrumentation, HMI/SCADA, networks, testing, troubleshooting and lifecycle documentation.

BUILD WINDOW	LABEL
8 February - 31 March 2027	Simulation / Training Project

COMPETENCIES	SYSTEM SCOPE
☐ Requirements and architecture ☐ Reusable equipment modules ☐ Process sequences and interlocks ☐ Analog instruments and validation ☐ SCADA and networks ☐ FAT and defect management ☐ Troubleshooting and handover	☐ Raw-water intake, transfer pumping, treatment tank, filtration and storage ☐ Pumps, valves, level/flow/pressure/quality signals ☐ Local protection retained if SCADA fails ☐ Clear Auto/Manual and Local/Remote ownership ☐ Alarm priorities, trends, quality indication and recovery ☐ Controlled scope: simulation only, with explicit exclusions

Acceptance criteria

- ☐ The system starts only when written permissives are true.
- ☐ Mode and command ownership are visible and deterministic.
- ☐ Normal operation, stop, fault and recovery match the control narrative.
- ☐ Bad signal or communications quality is visible and cannot look normal.
- ☐ Each critical fault produces a defined safe response and useful diagnostic.
- ☐ Documents, PLC/HMI tags and test records remain traceable by revision.

Project 04 build plan

Window	Stage	Required output
8-14 Feb	Baseline	URS-lite, P&ID, architecture, I/O, instruments and acceptance criteria
15-21 Feb	Integrate	Equipment modules, sequences, HMI/SCADA and communications
22-28 Feb	FAT	Normal, mode, fault, network-loss and recovery tests
1-14 Mar	Harden	Fix defects, run regression, audit tags and backups
15-31 Mar	Present	Final technical demonstration, evidence index and honest limitations

REQUIRED FAULT INJECTIONS

- ☐ Pump start failure
- ☐ Level transmitter open loop
- ☐ Valve position discrepancy
- ☐ Filter high differential pressure
- ☐ PLC-SCADA network loss
- ☐ Stale historian value
- ☐ Incorrect mode ownership
- ☐ Alarm flood after restart
- ☐ Wrong tag scale
- ☐ Backup/restore mismatch

EVIDENCE PACKAGE

- ☐ Requirements and scope
- ☐ P&ID and system architecture
- ☐ I/O, instrument, alarm and tag registers
- ☐ Control narrative and cause-and-effect
- ☐ PLC/HMI/SCADA evidence
- ☐ Network plan
- ☐ FAT and defect log
- ☐ Fault reports
- ☐ Backup/change log
- ☐ Five-to-ten-minute demo

Project 04 demonstration and scoring

Demonstration script

1. Open with requirements and safe-state philosophy
2. Trace one critical loop end to end
3. Run normal automatic operation
4. Inject network loss and prove local protection
5. Show defect, fix and regression evidence
6. Close with limitations and next engineering steps

Golden differentiators

- ☐ Requirements-to-test traceability matrix
- ☐ One-page executive architecture
- ☐ Network-failure philosophy
- ☐ Defect closure chart
- ☐ Short narrated demo with timestamps

Category	Weight	Release question
Correct function	25%	Does normal operation match the written narrative?
Fault behavior	25%	Are faults isolated, protected and recoverable?
Traceability	20%	Do drawings, tags, logic and tests agree?
Operator clarity	15%	Are mode, quality, alarms and actions obvious?
Explanation	15%	Can the work be explained honestly without notes?

RELEASE GATE

Target at least 80/100, with no category below 60 percent. A safety, ownership or data-quality defect blocks release even if the total score passes.

Portfolio file structure and evidence index

Use the same predictable structure for all four projects.

RECOMMENDED FOLDER PATTERN

00_README | 01_Requirements | 02_Design | 03_PLC | 04_HMI_SCADA | 05_Network | 06_Testing | 07_Faults | 08_Demo | 09_Backups

Evidence index fields

- ☐ Evidence ID
- ☐ Project and document title
- ☐ Revision and date
- ☐ Studied / Simulated / Implemented / Tested / Documented
- ☐ What the artifact proves
- ☐ Tool or platform used
- ☐ Linked requirement or test
- ☐ Known limitation
- ☐ File path

README opening statement

This is a personal Simulation / Training Project created to demonstrate junior automation and control-system skills. It was not commissioned on a live plant. Safety functions, field wiring and site behavior are represented only to the extent stated in the scope and limitations.

FINAL PRESENTATION STANDARD

Lead with the process problem, show the architecture, prove one normal cycle, inject one fault, explain the diagnosis, show recovery and close with documented limitations. Keep evidence stronger than adjectives.